

Paolo is saving for a trip to the Bahamas with his friends. He has estimated that his travel expenses will cost between \$2,000 and \$3,000 total. He has \$250 in savings, and he earns \$15.25 per hour at his job. He wants to know how many hours he must work in order to afford the trip.

1. Write an inequality to represent the situation.

$$2,000 \leq 250 + 15.25x \leq 3,000$$

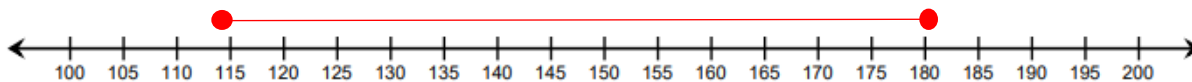
2. Solve the inequality.

$$\begin{array}{rcl} 2,000 & \leq & 250 + 15.25x \leq 3,000 \\ -250 & -250 & -250 \end{array}$$

$$\frac{1750}{15.25} \leq \frac{15.25x}{15.25} \leq \frac{2750}{15.25}$$

$$114.75 \leq x \leq 180.33$$

3. Graph the solution.



4. Check any that apply.

- ☒ Solutions must be non-negative.
☐ Solutions must be integers.

5. Explain your selection(s) from question 4.

Paolo can work fractions of hours but can't work negative hours.

Celine just started her own business making cake pops. She sells the cake pops in boxes, charging \$20.00 per box. It costs her \$65.00 to buy the supplies to make the cake pops. She wants to calculate how many boxes she must make to earn a profit of at least \$225.00.

6. Write an inequality to represent the situation.

$$20x - 65 \geq 225$$

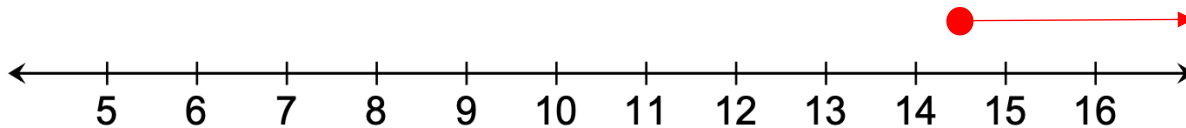
7. Solve the inequality.

$$\begin{array}{r} 20x - 65 \geq 225 \\ +65 \quad +65 \end{array}$$

$$\frac{20x}{20} \geq \frac{290}{20}$$

$$x \geq 14.5$$

8. Graph the solution.



9. Check any that apply.

- ☒ Solutions must be non-negative.
☒ Solutions must be integers.

10. Explain your selection(s) from question 9.

Celine can't sell negative boxes and she can't sell fractional parts of boxes.